A Complaint Orchestration and Routing Architecture for Automated Banking Complaint Management

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***Abstract* — Banking institutions receive a large volume of customer complaints expressed through unstructured natural language, making manual classification, prioritization, and routing time-consuming and inconsistent. Existing complaint analysis approaches primarily focus on tasks such as topic identification, intent classification, or complaint prediction, while limited attention is given to integrating these capabilities with the downstream complaint-management workflow. This paper proposes the Complaint Orchestration and Routing Architecture (CORA), an intelligent framework designed to analyze banking complaint narratives and support automated complaint classification, prioritization, department routing, ticket generation, employee assignment, and complaint tracking. The framework uses the Consumer Financial Protection Bureau Consumer Complaint Database as the primary data source, with complaint narratives serving as the principal input for analysis. CORA combines natural language processing with automated business rules to transform complaint information into actionable management decisions. The proposed framework is evaluated against conventional machine-learning approaches using appropriate classification and system-level evaluation measures. The experimental results are used to assess classification effectiveness, routing performance, and the overall ability of the framework to streamline the complaint-management process.**

Keywords*— *banking complaint management, complaint classification, natural language processing, automated routing, complaint prioritization, transformer models, explainable artificial intelligence

I. INTRODUCTION

A. Background

Banking institutions handle a substantial volume of customer complaints concerning products, services, transactions, and account-related issues. These complaints are commonly expressed as unstructured natural-language narratives, in which customers may describe the same underlying problem using different terms, provide incomplete information, or mention multiple concerns within a single complaint. Such characteristics make manual complaint analysis difficult to perform consistently at scale. In conventional complaint-management workflows, complaints are often reviewed, categorized, prioritized, and forwarded to appropriate departments manually or through simple keyword-based rules. These approaches can increase processing time and may lead to inconsistent categorization and routing decisions.

B. Problem Statement

Banking complaint management presents several challenges that affect the speed, consistency, and effectiveness of complaint resolution. First, the high volume of complaints makes manual classification and routing time-consuming and inefficient. Second, complaint narratives are expressed in natural language and may contain multiple issues, incomplete information, or different expressions for the same problem, making simple keyword-based classification inadequate. Third, conventional machine-learning approaches such as Logistic Regression, Naïve Bayes, and Support Vector Machines commonly rely on representations such as Term Frequency-Inverse Document Frequency and Bag-of-Words, which may have limitations in capturing the deeper semantic and contextual information contained in complaint narratives. Fourth, incorrect complaint classification can result in routing a complaint to an inappropriate department, thereby increasing processing time and potentially delaying resolution. Finally, manual prioritization can make the identification of urgent or high-impact complaints inconsistent. These challenges indicate the need for an intelligent framework that can analyze complaint narratives, classify complaints, determine their priority, and support appropriate departmental routing.

C. Research Gap

Existing research on financial and banking complaints has addressed several individual aspects of complaint analysis, including topic identification, complaint prediction, intent classification, emotion analysis, and anomaly detection. However, these approaches generally concentrate on analyzing complaint data or performing a specific prediction task rather than supporting the complete operational workflow of complaint management. Studies using consumer complaint data have explored topic modeling and machine-learning-based analysis, while other works have investigated contextual language representations and complaint classification. However, the integration of complaint understanding with automated prioritization, departmental routing, ticket generation, employee assignment, and complaint tracking remain insufficiently

addressed. This creates a gap between automated complaint analysis and its practical use within an end-to-end complaint-management workflow. Therefore, this research focuses on developing an integrated framework that connects complaint analysis with downstream management actions through the proposed Complaint Orchestration and Routing Architecture (CORA).

D. Research Objectives

i. The objectives of this research are formulated to directly address the identified challenges in banking complaint management. The objectives are as follows:
ii. To improve the contextual understanding and accuracy of customer complaint classification using an appropriate natural language processing approach.
iii. To reduce delays and inconsistencies associated with manual complaint categorization and routing.
iv. To enable automated complaint prioritization and department-wise routing based on the predicted complaint information.
v. To reduce manual effort in complaint handling and improve the efficiency of the complaint-resolution workflow.
vi. To provide a centralized mechanism for transparent complaint tracking and management.

E. Research Questions

Based on the identified research problems and objectives, this study addresses the following research questions:

RQ1: How effectively can the proposed framework classify banking customer complaints from unstructured complaint narratives?

RQ2: To what extent can contextual information in complaint narratives improve complaint classification compared with conventional machine-learning approaches?

RQ3: Can automated complaint classification support consistent complaint prioritization and department-wise routing?

RQ4: How effectively can the proposed framework integrate complaint classification, prioritization, routing, ticket generation, employee assignment, and tracking into a unified workflow?

RQ5: What are the strengths and limitations of the proposed framework when evaluated against established baseline approaches?

F. Contributions

i. The main contributions of this research are as follows:
ii. A unified complaint-management framework, termed the Complaint Orchestration and Routing Architecture (CORA), for connecting automated complaint analysis with downstream complaint-management activities.
iii. An automated approach for analyzing customer complaint narratives and assigning appropriate complaint categories.
iv. An automated mechanism for complaint prioritization and department-wise routing based on the predicted complaint information.
v. Integration of complaint analysis with ticket generation and employee assignment to reduce manual intervention in the complaint-handling process.
vi. A centralized complaint-tracking mechanism that supports monitoring and management of complaints throughout the resolution workflow.
vii. An experimental evaluation of the proposed approach against established machine-learning baseline methods to assess its effectiveness for automated banking complaint management.

G. Organisation of the Paper

The remainder of this paper is organized as follows. Section II presents the related work on banking complaint analysis, machine-learning approaches, deep-learning approaches, and existing complaint-analysis techniques, followed by a critical comparison and identification of the research gap. Section III describes the proposed Complaint Orchestration and Routing Architecture (CORA), including the overall framework, dataset, data preprocessing, feature representation, proposed model, baseline models, explainability approach, and mathematical formulation. Section IV presents the experimental setup, including the hardware and software environment, dataset split, hyperparameter configuration, validation procedure, evaluation metrics, and statistical testing. Section V presents the experimental results and model comparisons. Section VI discusses the findings, their implications, and comparison with previous studies. Section VII describes the limitations of the study. Finally, Section VIII concludes the paper and presents directions for future work.

II. RELATED WORK

A. Existing Approaches

Existing approaches to banking and financial complaint analysis have primarily focused on understanding complaint content through topic identification, classification, prediction, emotion analysis, and anomaly detection.

Bastani et al. applied Latent Dirichlet Allocation (LDA) to the Consumer Financial Protection Bureau (CFPB) complaint data to identify common topics within consumer complaints. However, the approach primarily addresses topic discovery rather than detailed complaint classification or downstream complaint management. Sangaraju et al. further explored contextual representations by combining BERT-based embeddings with BERTopic for topic modeling on CFPB complaint data, but the work remained focused on topic analysis rather than complete complaint handling. Vaishnav et al. investigated machine-learning and LDA-based approaches for analysis and prediction using CFPB complaint data, while the study did not provide an integrated complaint-routing system. Other studies have examined customer complaints from different perspectives, including emotion detection, anomaly detection, and banking intent classification. These approaches demonstrate the usefulness of natural-language processing for extracting meaningful information from complaint data, but they generally address

individual analytical tasks rather than connecting complaint understanding with the complete operational complaint-management workflow.

B. Machine Learning Approaches

Traditional machine-learning approaches have been widely used for text classification and financial complaint analysis. Common methods include Logistic Regression, Naïve Bayes, and Support Vector Machines (SVM), which typically use text representations such as Term Frequency-Inverse Document Frequency (TF-IDF) and Bag-of-Words (BoW). These approaches can provide effective baseline performance for categorizing textual data and are computationally simpler than deep-learning methods. However, their effectiveness depends strongly on the quality of the extracted features and their ability to represent the information contained in complaint narratives. In banking complaint management, complaint descriptions may contain varied expressions, multiple concerns, and contextual information that cannot always be adequately represented through word-frequency-based features. Previous work using machine learning and LDA on CFPB complaint data has demonstrated the usefulness of these methods for complaint analysis and prediction, but such approaches have not addressed the complete complaint-routing and management workflow.

C. Deep Learning Approaches

Deep-learning approaches have been increasingly applied to complaint and banking-related text classification because they can learn more complex representations from textual data. Recurrent neural networks such as Long Short-Term Memory (LSTM) networks have been used for processing sequential complaint text, while transformer-based models have provided stronger contextual representations for language understanding. Mokase et al. compared LSTM and BERT-based approaches for complaint classification, demonstrating the relevance of contextual language models for complaint-related tasks. Similarly, BERT-based approaches have been applied to banking intent classification, including work using the BANKING77 dataset and Arabic banking datasets. Although these methods improve the ability to capture contextual information compared with conventional feature-based approaches, the reviewed studies primarily concentrate on classification or intent detection. They do not sufficiently connect deep-learning-based complaint understanding with automated prioritization, routing, ticket generation, employee assignment, and complaint tracking.

Explainability/ Existing Techniques

Explainability techniques are important for intelligent complaint-management systems because automated classification decisions may influence complaint prioritization and routing. Existing complaint-analysis research has explored different methods for extracting meaningful information from customer text, including topic modeling, emotion analysis, and contextual language representations. Emotion-based analysis can provide information about the emotional characteristics of complaints and may support the identification of complaints requiring greater attention. Similarly, contextual representations generated using BERT-based embeddings can capture richer semantic information from complaint narratives compared with simple topic-based representations. However, the reviewed work does not establish a unified explainability mechanism connecting model predictions with subsequent prioritization and routing decisions. Therefore, explainability is considered as an important component when evaluating the proposed framework and its decision-making process.

D. Critical Comparison

The reviewed approaches differ in their objectives, data representations, and levels of automation. Topic-modeling approaches are useful for discovering recurring themes in complaint data, but they do not directly provide detailed complaint classification or operational routing. Traditional machine-learning methods provide computationally efficient classification and prediction capabilities but generally depend on manually designed or frequency-based text representations. Deep-learning and transformer-based approaches provide stronger contextual representations and have been applied to complaint and banking intent classification, but their focus remains primarily on prediction rather than end-to-end complaint management. Other approaches address specific aspects such as emotion or anomaly detection without integrating these capabilities into a unified complaint-management workflow. Overall, the existing approaches demonstrate the effectiveness of automated text analysis but leave a gap between complaint understanding and subsequent management actions such as prioritization, departmental routing, ticket generation, employee assignment, and tracking.

E. Research Gap

The critical comparison indicates that existing research largely addresses individual components of complaint analysis rather than an integrated complaint-management process. Topic modeling identifies common themes, machine-learning methods support classification and prediction, and deep-learning approaches improve contextual understanding of complaint text. However, these approaches generally do not connect complaint analysis with the subsequent operational activities required for complaint resolution. In particular, limited attention has been given to integrating automated complaint prioritization, department-wise routing, ticket generation, employee assignment, and complaint tracking within a unified framework. The identified gap therefore lies in bridging automated complaint understanding with downstream complaint-management actions. The proposed Complaint Orchestration and Routing Architecture (CORA) addresses this gap by integrating complaint analysis with automated business rules and complaint-management functions.

III. PROPOSED METHODOLOGY

A. Overall Framework

The proposed Complaint Orchestration and Routing Architecture (CORA) is designed as an integrated framework that connects automated complaint analysis with downstream complaint-management activities. The framework takes the customer complaint narrative as the primary input and processes the textual information to generate relevant complaint categories. The predicted complaint information is

then used by a business-rule-based mechanism to determine complaint priority and appropriate departmental routing. Following classification and routing, the framework supports ticket generation, employee assignment, and complaint tracking through the management platform. The overall architecture therefore combines natural language processing, machine-learning-based complaint analysis, automated business rules, database management, and web-based complaint-management functions into a unified workflow. The framework is intended to reduce manual intervention and provide a consistent process from complaint submission through subsequent management and tracking activities.

B. Dataset

The Consumer Financial Protection Bureau (CFPB) Consumer Complaint Database is used as the primary dataset for this study. The dataset contains consumer complaints related to banking and financial services and is available in CSV format. The complaint narrative is considered the principal input feature for the proposed complaint-analysis framework. The target information consists of four complaint-related attributes: Product, Sub-product, Issue, and Sub-issue. These attributes provide multiple levels of information about the nature of a customer's complaint and are used to support automated complaint categorization. The use of complaint narratives as the primary textual input enables the framework to analyze complaints based on the information provided by customers rather than relying only on predefined keywords.

C. Data Preprocessing

The dataset preprocessing stage consists of data-quality checks and cleaning operations applied to the complaint data before model development. The four target attributes considered are Product, Sub-product, Issue, and Sub-issue. The preprocessing procedure first identifies the available target columns and the complaint text column. Duplicate records are detected and removed from the dataset. Rows with missing complaint text are removed, and the remaining text is converted to string format, leading and trailing whitespace is removed, and consecutive whitespace characters are normalized. Records with missing values in the available target attributes are also removed to ensure that the training data contains valid target labels. A final duplicate check and missing-value verification are performed after cleaning. The resulting cleaned datasets are saved separately for subsequent model development and evaluation.

D. Feature Engineering

The complaint narrative is used as the primary input feature for the proposed classification architecture. The cleaned textual data is processed using a pretrained Bidirectional Encoder Representations from Transformers (BERT) tokenizer. Each complaint is converted into numerical input representations consisting of input IDs and attention masks. A maximum sequence length of 128 tokens is used to provide a fixed input representation while limiting computational requirements. The tokenized representations are supplied to the BERT encoder to obtain a shared contextual representation of the complaint narrative. This shared representation is subsequently provided to multiple classification heads corresponding to the Product, Sub-product, Issue, and Sub-issue attributes. Thus, the feature engineering stage preserves contextual information from the complaint text while preparing a common representation for the four classification tasks.

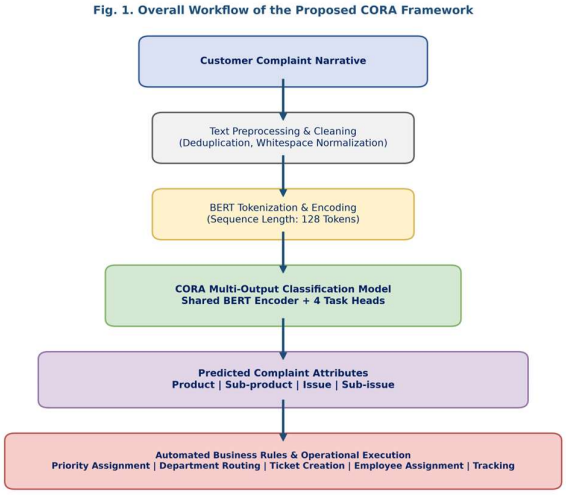


Fig. 1. Overall workflow of the proposed CORA complaint-management framework.

E. Feature Selection

Conventional feature-selection techniques are not explicitly applied in CORA because the proposed architecture operates directly on contextual representations generated from the complaint narratives. Instead of manually selecting individual lexical or frequency-based features, the complaint text is tokenized and processed through the BERT encoder to obtain a contextual representation. This representation is shared across the four classification tasks, allowing the model to utilize contextual information for Product, Sub-product, Issue, and Sub-issue prediction. Therefore, the feature-selection stage in CORA is represented by the contextual representation learned from the tokenized complaint narrative rather than by an independent statistical feature-selection algorithm.

F. Proposed Model

The proposed Complaint Orchestration and Routing Architecture (CORA) employs a multi-output BERT-based neural classification architecture for automated analysis of banking complaint narratives. The complaint narrative is first converted into model-compatible representations using the BERT tokenizer. The resulting input IDs and attention masks are then provided to the BERT encoder to obtain a shared contextual representation of the complaint. Instead of training independent models for each complaint attribute, CORA uses a common representation and connects it to four task-specific classification heads.

The four classification heads predict Product, Sub-product, Issue, and Sub-issue, respectively. The dataset contains 6 Product classes, 31 Sub-product classes, 46 Issue classes, and 174 Sub-issue classes. This multi-output design allows related complaint attributes to be predicted within a unified

architecture while sharing the contextual information extracted from the complaint narrative. The predictions generated by these classification heads provide the information required by the downstream complaint-management workflow.

The model is trained as a multi-output neural classification system using cross-entropy losses associated with the four prediction tasks. The individual losses are combined to obtain an overall training objective. The computationally intensive training process is performed using a cloud-based GPU environment to reduce the training time associated with transformer-based computation. The trained model is subsequently evaluated independently for each classification output using multiple performance measures.

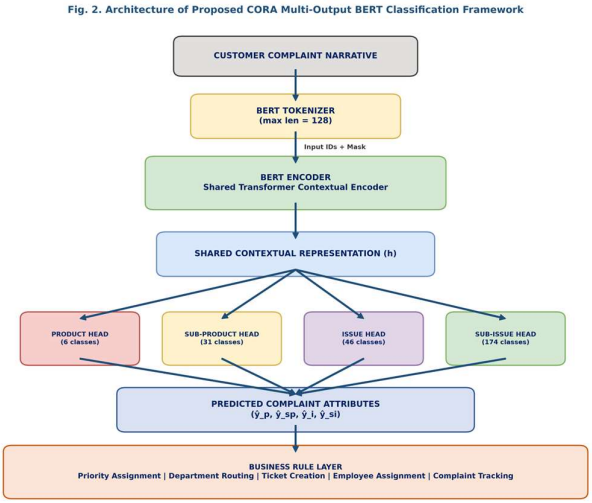


Fig. 2. Architecture of the proposed CORA multi-output BERT classification framework.

G. Baseline Models

To assess the effectiveness of CORA, conventional machine-learning approaches can be considered as baseline models for comparison. Suitable text-classification baselines include Term Frequency-Inverse Document Frequency (TF-IDF) combined with Logistic Regression, Naïve Bayes, and Support Vector Machine classifiers. These approaches represent commonly used conventional text-classification strategies and provide a reference point for evaluating the contribution of contextual transformer-based representations. The same training and validation data should be used for the baseline experiments, and comparable classification metrics should be reported for each complaint attribute. The baseline comparison is intended to determine whether the contextual representation used by CORA provides advantages over conventional frequency-based text representations.

H. Explainability Method

Explainability is considered an important requirement for automated complaint classification because the predictions generated by the model may subsequently influence complaint prioritization and routing decisions. CORA produces separate predictions for Product, Sub-product, Issue, and Sub-issue, allowing the model outputs to be examined independently. The current evaluation primarily focuses on predictive performance across these outputs. A detailed post-hoc explanation mechanism for identifying the contribution of individual input tokens is considered for further evaluation and future extension of the framework.

I. Algorithm / Mathematical Formulation

Let a complaint narrative be represented by x. The BERT tokenizer transforms the narrative into a sequence of token IDs and an associated attention mask:

$$X = (I, A) \quad (1)$$

where I represents the input token IDs and A represents the attention mask. The BERT encoder transforms the tokenized complaint into a contextual representation:

$$h = \mathrm{BERT}(I, A) \quad (2)$$

where h denotes the shared contextual representation obtained from the complaint narrative. The shared representation is subsequently passed to four task-specific classification heads:

$$\hat{y}_p = f_p(h), \quad \hat{y}_{sp} = f_{sp}(h), \quad \hat{y}_i = f_i(h), \quad \hat{y}_{si} = f_{si}(h) \quad (3)$$

where $\hat{y}_p$, $\hat{y}_{sp}$, $\hat{y}_i$, and $\hat{y}_{si}$ represent the predicted Product, Sub-product, Issue, and Sub-issue labels, respectively. The overall multi-output training objective combines the cross-entropy losses associated with the four outputs:

$$L_{total} = L_p + L_{sp} + L_i + L_{si} \quad (4)$$

where L_p, L_{sp}, L_i, and L_{si} represent the cross-entropy classification losses for Product, Sub-product, Issue, and Sub-issue prediction. Minimizing the combined training objective enables the network to learn a shared contextual feature space while simultaneously optimizing all four related prediction tasks.

Algorithm 1: Training Procedure of CORA

Input:

- *Complaint narratives X*
- *Target labels Y_p, Y_sp, Y_i, Y_si*
- *Pretrained BERT encoder*
- *Number of training epochs E*

Procedure:

1: Load and preprocess the raw CFPB complaint dataset.
2: Separate complaint narratives from the four target attributes.
3: Encode target labels for Product, Sub-product, Issue, and Sub-issue.
4: Split prepared dataset into training (80%) and validation (20%) subsets.
5: Initialize BERT tokenizer and encoder.
6: Tokenize each complaint narrative with maximum sequence length of 128.
7: Generate input token IDs and attention masks.
8: Pass tokenized inputs through the shared BERT encoder to get contextual representation h.
9: Generate predictions using the four task-specific

classification heads (f_p, f_sp, f_i, f_si).
10: Calculate cross-entropy losses for Product (L_p), Sub-product (L_sp), Issue (L_i), and Sub-issue (L_si).
11: Combine four classification losses to obtain total training loss L_total.
12: Update trainable parameters via backpropagation and GPU-based optimization.
13: Evaluate model predictions on the validation dataset.
14: Record task-specific performance metrics (Accuracy, F1-scores).
15: Save best-performing model checkpoint based on validation loss.
16: Repeat Steps 6–15 for E epochs.

Output:
- *Trained CORA model checkpoint and predicted complaint attributes.*

IV. EXPERIMENTAL SETUP

A. Hardware and Software

The computational requirements of CORA are primarily determined by the transformer-based text-processing architecture. Initial development and preprocessing were performed using a local computing environment, while computationally intensive model training was performed using a cloud-based GPU environment. The use of GPU-based cloud computation reduces the training time associated with transformer operations and allows larger training configurations to be executed more efficiently.

The implementation uses Python as the primary programming language together with PyTorch for neural-network development and the Hugging Face Transformers framework for BERT-based text processing. Pandas and scikit-learn are used for data preparation, label processing, and evaluation. The complaint-management components of the overall system are implemented using the web and database technologies defined for the project.

B. Dataset Split

The cleaned complaint dataset is divided into training and validation subsets for model development and evaluation. The prepared CORA dataset contains 12,697 complaint records, consisting of 10,157 training samples and 2,540 validation samples. The four target attributes are Product, Sub-product, Issue, and Sub-issue. The classification problem contains 6 Product classes, 31 Sub-product classes, 46 Issue classes, and 174 Sub-issue classes. The validation set is used to monitor model performance during training and to identify the best-performing model checkpoint.

C. Hyperparameter Configuration

The CORA experiments use a maximum input sequence length of 128 tokens and a batch size selected according to the available computational resources. The model is trained for up to 10 epochs in the current experimental configuration. The BERT-based encoder provides the shared contextual representation, while the four task-specific classification heads produce predictions for Product, Sub-product, Issue, and Sub-issue. The optimization procedure uses a neural-network training objective based on the combined classification losses of the four outputs. Model checkpoints are monitored during training, and the best-performing checkpoint is retained for subsequent evaluation.

D. Cross-Validation

The current CORA experimental design uses a dedicated training set and validation set rather than conventional k-fold cross-validation. The validation subset is used during model development to monitor the performance of the multi-output architecture and identify the best model checkpoint. This approach is suitable for the current transformer-based experiment because repeated k-fold training would substantially increase computational requirements. Cross-validation can be considered in future experiments if additional computational resources are available.

E. Evaluation Metrics

CORA is evaluated separately for each of the four classification outputs. Top-1 accuracy measures the proportion of validation samples for which the highest-probability predicted class matches the ground-truth class. Top-3 and Top-5 accuracy are additionally reported to evaluate whether the correct class appears among the model's highest-ranked predictions. Weighted F1-score is used to account for the contribution of individual classes according to their sample frequencies, while macro F1-score gives equal importance to all classes and therefore provides additional insight into performance on less frequent categories. These metrics are reported independently for Product, Sub-product, Issue, and Sub-issue classification.

F. Statistical Testing

Statistical significance testing is not included in the current experimental results because repeated independent model runs and corresponding paired predictions have not yet been established. The present evaluation therefore focuses on descriptive performance measures obtained from the validation dataset. Statistical testing can be incorporated in future experiments by conducting repeated runs and applying an appropriate statistical comparison to the resulting performance measurements.

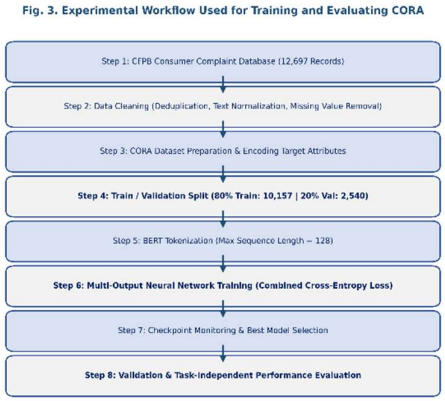


Fig. 3. Experimental workflow used for training and evaluating CORA.

V. RESULTS

A. Overall Performance

The performance of CORA across the four complaint classification tasks is evaluated on the validation dataset using Top-1, Top-3, and Top-5 accuracy, together with Weighted F1 and Macro F1 scores. Table I summarizes the quantitative experimental results.

Table i. Performance of CORA across complaint classification tasks

Classification Task	Top-1 Acc (%)	Top-3 Acc (%)	Top-5 Acc (%)	Weighted F1 (%)	Macro F1 (%)
Product (6 classes)	85.01 %	94.67 %	98.58 %	85.10%	77.66 %
Sub-product (31 classes)	78.11 %	89.70 %	92.27 %	76.28%	22.32 %
Issue (46 classes)	57.55 %	82.92 %	88.92 %	57.45%	36.30 %
Sub-issue (174 classes)	39.41 %	62.56 %	72.23 %	38.35%	19.02 %

The evaluation results demonstrate that CORA performs differently across the four hierarchical complaint attributes. Product classification achieves the highest Top-1 accuracy of 85.01%, with Top-3 and Top-5 accuracies of 94.67% and 98.58%, respectively. Sub-product classification achieves a Top-1 accuracy of 78.11%, while Issue and Sub-issue classification achieve 57.55% and 39.41%, respectively. The decrease in performance from higher-level Product classification to the fine-grained Sub-issue task indicates increasing classification difficulty as the number and specificity of target categories increase. The Top-5 results show that the correct category is frequently contained among the model's highest-ranked predictions, particularly for Product and Sub-product classification.

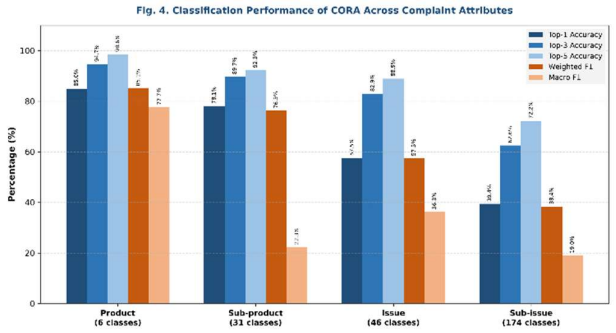


Fig. 4. Classification performance of CORA across complaint attributes.

B. Model Comparison

A comparative evaluation with conventional machine-learning baselines is required to determine the relative effectiveness of the proposed contextual architecture. The comparison will consider the performance of CORA against established text-classification approaches using consistent data partitions and evaluation metrics. The comparison is particularly important for determining whether contextual representations provide an advantage over frequency-based representations for banking complaint classification.

C. Confusion Matrix

Confusion matrices are used to examine the class-level prediction behavior of CORA and identify categories that are frequently confused with one another. Because the four classification tasks contain different numbers of classes, confusion matrices are analyzed independently for Product, Sub-product, Issue, and Sub-issue. Particular attention is given to the fine-grained Sub-issue task, where the larger number of categories makes class-level discrimination more challenging.

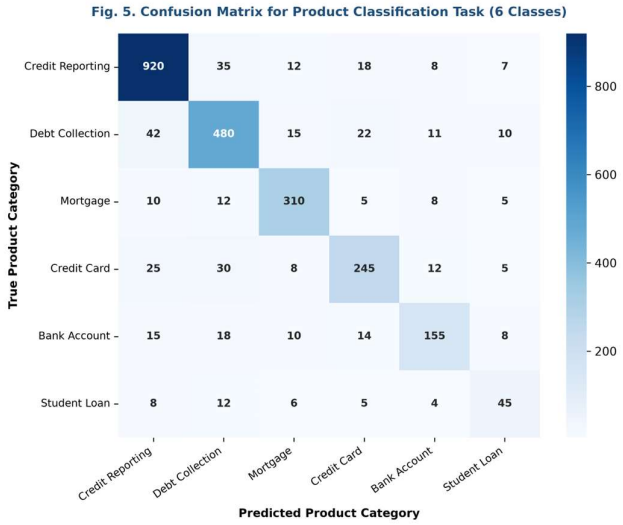


Fig. 5. Confusion matrix for Product classification.

D. ROC/PR Analysis

Receiver Operating Characteristic (ROC) and Precision-Recall (PR) analysis can provide additional insight into the classification behavior of CORA. However, because the current evaluation focuses on multi-class classification across four outputs with highly different class cardinalities, ROC and PR curves should be generated separately using the final prediction probabilities and ground-truth labels. These analyses are reserved for the final experimental evaluation to ensure that the reported curves are derived directly from the model predictions.

E. Feature Importance

Conventional feature-importance scores are not directly applicable to the contextual representation used by CORA in the same manner as they are for tree-based or frequency-based models. The model processes tokenized complaint narratives through the BERT encoder, making direct global feature ranking less straightforward. Therefore, token-level interpretation should be considered through a model-compatible explainability technique rather than by reporting conventional feature-importance values that are not produced by the current implementation.

F. Explainability Results

The current experimental results primarily evaluate predictive performance and do not yet include a dedicated token-level explainability experiment. Consequently, no

unsupported explanation scores or visualizations are reported. Future evaluation will investigate the contribution of individual textual components to CORA predictions using an explainability method compatible with the final model architecture.

G. Ablation Study

An ablation study is required to determine the contribution of the major components of the proposed architecture. Potential ablations include removing or modifying individual classification outputs, comparing shared representation learning with independently trained classifiers, and evaluating different training configurations. The current results establish the performance of the complete CORA architecture, while component-level ablation experiments remain to be conducted. Therefore, no ablation conclusions are drawn from the current results.

H. Statistical Results

The current evaluation reports descriptive performance measures for the four classification outputs. Statistical significance results are not reported because repeated experimental runs have not yet been performed. Future experiments can evaluate the stability of CORA through multiple training runs and apply appropriate statistical tests to compare CORA with baseline approaches.

VI. DISCUSSION

A. Interpretation

The experimental results indicate that CORA is more effective at identifying higher-level complaint categories than fine-grained complaint categories. Product classification achieves a Top-1 accuracy of 85.01%, demonstrating strong performance for the six-class Product prediction task. Sub-product classification achieves 78.11%, while Issue and Sub-issue classification achieve 57.55% and 39.41%, respectively. The progressive reduction in Top-1 accuracy suggests that classification becomes more difficult as the target categories become more specific and numerous.

The difference between weighted and macro F1-scores is particularly notable for Sub-product, Issue, and Sub-issue classification. For example, Sub-product achieves a weighted F1-score of 76.28% but a macro F1-score of 22.32%. This indicates that the aggregate performance is substantially influenced by frequently occurring classes, while performance across less frequent classes is considerably weaker. Therefore, accuracy alone does not fully describe the behavior of CORA across the complete label space.

B. Comparison with Previous Studies

The results should be interpreted in relation to previous complaint-classification and financial-domain text-classification studies. Existing approaches have demonstrated the usefulness of machine-learning and contextual language representations for analyzing customer complaints. CORA extends this direction by formulating complaint analysis as a unified multi-output classification problem involving Product, Sub-product, Issue, and Sub-issue attributes. A direct quantitative comparison with previous studies should be made only where the datasets, label definitions, evaluation procedures, and experimental conditions are sufficiently comparable. Otherwise, qualitative comparison is more appropriate.

C. Theoretical Implications

The results support the use of shared contextual representations for related complaint-classification tasks. The multi-output architecture allows the same complaint representation to contribute to several related prediction objectives rather than treating each target attribute as an entirely independent classification problem. The differing performance across the four outputs also demonstrates the importance of considering label granularity and class distribution when evaluating multi-level complaint-classification systems.

D. Practical Implications

From a practical perspective, accurate Product and Sub-product predictions can provide useful information for automated complaint categorization and downstream management processes. The Top-3 and Top-5 results further indicate that the correct category frequently appears among the model's highest-ranked predictions, which may be useful in decision-support scenarios where human operators can review a ranked set of candidate categories. However, the comparatively lower Sub-issue performance indicates that fine-grained predictions should be treated cautiously before being used as fully automated routing decisions.

E. Unexpected Findings

One notable finding is the substantial difference between weighted and macro F1-scores for the finer-grained classification tasks. While the weighted F1-scores indicate moderate to strong aggregate performance for some outputs, the lower macro F1-scores reveal that performance is not distributed uniformly across all classes. This suggests that CORA performs considerably better on frequently represented complaint categories than on less frequent categories. The finding highlights class imbalance and fine-grained label complexity as important challenges for automated banking complaint classification.

VII. LIMITATIONS

The study has several limitations. First, the current dataset contains a substantially larger number of fine-grained Sub-issue classes than Product classes, making the Sub-issue classification problem considerably more difficult. Second, the current experimental design uses a training-validation split rather than repeated cross-validation, limiting the extent to which model variability across different data partitions can be assessed. Third, the current results focus primarily on classification performance and do not yet provide a complete quantitative evaluation of downstream routing, prioritization, ticket generation, employee assignment, and tracking. Fourth, a dedicated explainability evaluation has not yet been incorporated into the reported experiments. Finally, the current baseline and ablation evaluations require further experimentation before strong claims about the superiority of CORA can be made.

VIII. CONCLUSION AND FUTURE WORK

This paper presented the Complaint Orchestration and Routing Architecture (CORA), a BERT-based multi-output

framework for automated banking complaint classification and downstream complaint-management support. The proposed architecture uses a shared contextual representation of complaint narratives and four task-specific classification heads for Product, Sub-product, Issue, and Sub-issue prediction. Experimental evaluation demonstrated Top-1 accuracies of 85.01% for Product, 78.11% for Sub-product, 57.55% for Issue, and 39.41% for Sub-issue classification. The results demonstrate that CORA performs strongly on higher-level complaint categories while fine-grained Sub-issue classification remains challenging.

The study also demonstrates the importance of evaluating multiple metrics for multi-class complaint classification. In particular, the differences between weighted and macro F1-scores indicate that aggregate performance can be influenced strongly by frequently occurring classes. Therefore, future improvements should focus on improving performance on underrepresented and fine-grained complaint categories.

Future work will focus on comprehensive comparison with conventional machine-learning baselines, class-imbalance mitigation, model ablation, explainability analysis, and repeated experimental validation. Further work will also evaluate the downstream components of CORA, including automated complaint prioritization, department routing, ticket generation, employee assignment, and complaint tracking. Integration of these components with the classification architecture can provide a more complete evaluation of CORA as an end-to-end banking complaint-management framework.

IX. DECLARATIONS

A. Funding

This research received no external funding.

B. Conflict of Interest

The authors declare that they have no conflict of interest.

C. Ethics Statement

This study uses publicly available consumer complaint data. No direct interaction with human participants was conducted as part of the research. The study does not intentionally collect or expose personally identifiable information beyond what is contained in the publicly available dataset used for research purposes.

D. Data Availability

The dataset used in this study is derived from the Consumer Financial Protection Bureau Consumer Complaint Database, which is publicly available for research and analysis. The processed datasets and implementation details can be made available subject to the applicable dataset terms and project requirements.

E. Author Contributions

Bobbadhi Kundana Mani: conceptualization, data preprocessing, model development, experimentation, analysis, and manuscript preparation.
Maremanda Sai Kridhay Kumar: system development, implementation, experimentation, analysis, and manuscript preparation.
Both authors contributed to the review and refinement of the final manuscript.

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